

Yi Tao

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Research interests

Artificial Intelligence in Education (AIED); human–AI interaction in teaching and learning; LLM-based pedagogical agents; AI-supported online discussion and collaborative knowledge construction.

Education

South China Normal University
Master's Student in Educational Technology
Sep. 2024 – Jun. 2027
GPA: 3.88/5.00

Shandong Normal University
Bachelor's Degree in Educational Technology
Sep. 2020 – Jun. 2024
GPA: 3.79/5.00

Publications and Manuscripts

Manuscripts Under Review

1. **Tao, Y.**, Li, H., Yin, L., Lu, J., & Zhang, Y. AI teaching assistants in asynchronous discussion forums: A mixed-methods study of students' Community of Inquiry presence. [Manuscript under review]
2. **Tao, Y.**, Wang, Z., Miao, J., Yang, B., Lai, R., & Dong, J. When users invoke AI in public discussions: Interactional patterns and depth of AI-mediated knowledge construction on social media. [Conference paper under review]
3. Miao, J., **Tao, Y.**, & Lin, X.-F. Opening the black box of logical remainders: A transparency reporting framework for fsQCA intermediate solutions. [Manuscript under review]

Journal articles

1. Hu, X., **Tao, Y.**, & Mao, J. (2026). Beyond human-centeredness: The posthuman turn in the conception of teaching subjects in the intelligent era. *Distance Education in China*, 46(5), 37–50.
<https://doi.org/10.13541/j.cnki.chinade.2026.05.009>. [CSSCI-indexed; in Chinese]
2. Hu, X., **Tao, Y.**, Xiao, D., et al. (2026). The actual patterns and configurational pathways of digital transformation in primary and secondary schools: A grounded theory and fuzzy-set qualitative comparative analysis based on 58 smart education model schools. *The Chinese Journal of ICT in Education*, 32(5), 38–48. [In Chinese]

3. **Tao, Y.**, Liu, J., Wang, S., et al. (2024). Characteristics and influencing mechanisms of learning engagement and learning gains among publicly funded teacher education students. *Western China Quality Education*, 10(4), 166–170. <https://doi.org/10.16681/j.cnki.wcqe.202404038>. [In Chinese]
4. Wang, Q., **Tao, Y.**, Fu, Y., et al. (2023). Exploring a primary and secondary school artificial intelligence teaching model oriented toward AI literacy under the STEAM concept. *Computer Education*, (11), 176–179, 185. <https://doi.org/10.16512/j.cnki.jsjy.2023.11.019>. [In Chinese]

Conference papers

1. Lu, J., Xie, Y., **Tao, Y.**, Lin, Y., Zhong, J., & Wang, Y. (2026). Audience-based education: Rethinking the logic of adaptive learning systems enhanced by large language models. In *Proceedings of the 2026 8th International Conference on Computer Science and Technologies in Education* (pp. 105–110). IEEE.
2. **Tao, Y.**, Ni, L., Li, H., et al. (2026). Design and empirical study of teaching assistant agents for enhancing presence in discussion forums. Accepted conference paper, 30th Global Chinese Conference on Computers in Education, GCCCE 2026.
3. **Tao, Y.**, Lu, J., Lin, Y., et al. (2025). Smart learning navigation supported by formative assessment: Model construction and scenario application. Conference paper, 24th International Forum of Educational Technology, IFET 2025.

Skills

Research Design: Mixed-methods research, quasi-experimental design, survey research, semi-structured interviews, classroom observation.

Qualitative and Quantitative Analysis: QCA/fsQCA, grounded theory coding, content analysis, thematic analysis, ANOVA, t-tests, regression analysis, SNA, ENA.

Technical Skills: Python, R, SPSS, NVivo, LLM-based agent design, API-based system integration, basic front-end development.

Languages and Certificates: CET-6; High School Information Technology Teacher Qualification Certificate.

Professional experience

2025 | Intern, Evaluation of Excellent AI Curriculum Resources for Primary and Secondary Schools, Guangdong Provincial Department of Education.

2025 | Intern, Evaluation of Thematic Project-based Learning Cases for Primary and Secondary Schools, Guangdong Provincial Department of Education.

2024–2025 | Teaching Assistant, Advanced Training Program on “Generative AI Empowering Future Teachers,” South China Normal University.

2024 | Student Coordinator, Digital Leadership Training Program for Primary and Secondary School Principals, Institute of Educational Artificial Intelligence, South China Normal University.

2023 | Information Technology Intern Teacher, Shandong Experimental High School.

2021–2023 | Student Assistant, Academic Affairs Office, Shandong Normal University.

Honors and awards

National Second Prize, iTeach Digital Education Application Innovation Competition, 2021.

Provincial Third Prize, China Collegiate Computing Design Competition, Shandong Division, 2022.

Graduate Scholarship, South China Normal University, 2024–2025, 2025–2026.

Outstanding Student Scholarship, Shandong Normal University, 2020–2021, 2021–2022, 2022–2023.

Research experience

My research focuses on Artificial Intelligence in Education (AIED), particularly human–AI interaction, AI-supported discussion, and collaborative knowledge construction in online and blended learning environments.

Empirical Research on AI-Supported Interaction and Knowledge Construction

2025–present | Research Lead & System Developer

LLM-based AI Teaching Assistant Agent for Asynchronous Discussion Forums

- Led the full research cycle, including theoretical framing, instructional design, system development, platform deployment, empirical evaluation, and manuscript writing.
- Designed a CoI-informed AI teaching assistant for asynchronous discussion forums, targeting cognitive presence, social presence, and teaching presence.
- Developed and integrated an LLM-based teaching assistant system into an online course discussion platform through API-based implementation.
- Deployed the system across five university courses involving over 300 students.
- Conducted a 10-week quasi-experimental mixed-methods study comparing AI TA-supported and human TA-supported discussions.
- Collected and analyzed CoI questionnaires, forum transcripts, reflection posts, and semi-structured interviews.
- Identified a perception–behavior gap in AI-supported discussions: the AI TA enhanced students' perceived CoI presence, while its influence on deeper collaborative inquiry behaviors remained limited.
- Produced multiple research outputs, including a master's thesis, an accepted GCCCE 2026 conference paper, and an English journal manuscript under review.

2026–present | Research Lead, First Author & Corresponding Author

Public AI Invocation and AI-Mediated Knowledge Construction on Social Media

- Investigated how conversational AI agents are invoked and taken up in public social media discussions as part of human–AI–human knowledge construction.
- Analyzed 869 knowledge-related AI-invocation episodes from rehydrated GrokSet conversations using interactional pattern analysis and an adapted Interaction Analysis Model.
- Developed an episode-level coding framework to identify public AI invocation patterns and observable levels of knowledge construction.
- Used LLM-assisted screening and coding with double human verification to support efficient and reliable data analysis.
- Identified six interactional patterns and found that deeper knowledge construction emerged mainly through evidence testing, argumentative intervention, and post-response uptake.
- Prepared a conference paper currently under review.

Earlier related work: K-12 AI curriculum and intelligent literacy

Participated in the design of a STEAM-based teaching model for K-12 artificial intelligence courses.

Developed an AI curriculum practice case aimed at improving students' intelligent literacy, including AI knowledge, AI capabilities, and AI ethics.

Theoretical research on human-AI interaction in education

2025–2026 | Lead Student Researcher & Corresponding Author

Teaching Subjectivity and Human-AI Relationships in the Intelligent Era

- Led the conceptual development and manuscript drafting of a theoretical study on teaching subjectivity in the intelligent era.
- Drew on posthumanist perspectives to rethink anthropocentric assumptions in traditional teaching theory.
- Proposed that teaching subjectivity is shifting from human individuals to human-machine symbiotic networks, from static presupposition to dynamic becoming, and from hierarchical opposition to distributed collaboration.
- Discussed ethical risks in AI-supported teaching, including responsibility fragmentation, erosion of teaching autonomy, and marginalization of humanistic values.
- Proposed governance pathways such as meaningful human control, teachers' critical digital literacy, and the balance between instrumental rationality and value rationality.

2025–present | Research Assistant

Teaching Model Innovation Based on Artificial Intelligence Technology

National Education Science Major Project, VCA250013

Participated in a national major project on AI-supported teaching model innovation.

Contributed to research on teaching model transformation based on continuous accompaniment and cognitive twins.

Engaged in theoretical and practical exploration of how AI technologies reshape teaching structures, teacher roles, and learning support mechanisms.

Configurational Research on AI-Enabled School Digital Transformation

2025–present | Principal Investigator / Lead Student Researcher

School Digital Transformation: Current Status and Advancement Pathways

South China Normal University Seed Project, 25JXKA05

- Led a research project on the current status and advancement pathways of school digital transformation.
- Conducted research based on 58 provincial smart education sample schools in Guangdong Province.
- Integrated grounded theory and fuzzy-set Qualitative Comparative Analysis to examine practical patterns and configurational pathways of school digital transformation.
- Identified key factors and pathways of school digital transformation, including policy-driven, institution-driven, and comprehensive-driven transformation models.
- Coordinated research design, data analysis, and manuscript drafting.

2026–present | Co-Principal Investigator

Decoding County-level School Digital Transformation: Challenges, Causes, and Solutions

Guangdong Climbing Program Project, pdjh2026bg066

- Co-led a provincial-level student research project on county-level school digital transformation.
- Participated in research design, field investigation, project coordination, and academic writing.